Question 1: Parts Order Form

Original table and 1NF

Key notation: primary-key fields are **bold and underlined**.

OrderRecord (1NF)

customer Name	customer Number	customer Type	**order Date**	**order Time**	**employee**	**part Number**	part Name	part Type	cage Code	quantity Ordered	unit Price
Jeff Peterson	HG54587	Consumer	7/1/2024	10:30am	D. Harrison	10654	Float Control	Plumbing	G413	4	12
Jeff Peterson	HG54587	Consumer	7/1/2024	10:30am	D. Harrison	10456	Modulator	Electrical	H433	3	7
Jeff Peterson	HG54587	Consumer	7/1/2024	10:30am	D. Harrison	10776	Hose Assembly	Plumbing	G413	7	9
Jeff Peterson	HG54587	Consumer	7/1/2024	10:30am	D. Harrison	10657	Float Assembly	Plumbing	G413	5	10

The form lists four parts in one order. To put it in 1NF, I use one row for each part and repeat the customer and order details in each row. Each cell then contains one value. Dates use the original M/D/YYYY format.

a. Assumptions

1. Each order is handled by one employee. I assume the employee labels identify different employees and that an employee handles no more than one order at the same recorded date and time. An order can therefore be identified by (employee, orderDate, orderTime). Different employees may handle orders at the same time.

2. A customer number identifies one customer with one recorded name and type. Employees can serve any customer; customerNumber does **not** determine employee, and employee does **not** determine customerNumber.

3. Each part number identifies one part with a name, type, cage code and unit price. I assume its price stays the same across orders and customers. The sample does not establish that a part type determines a cage code, so I do not use that dependency.

4. A part occurs at most once per order; repeat quantities for that part are combined. quantityOrdered belongs to the order-part combination. All key values are present.

Field mapping: Date becomes orderDate; Time becomes orderTime; Name and Type in the part section become partName and partType. Other field labels are converted to lower camel case. These are renamings only; no attribute or surrogate key has been added.

1NF primary key: (employee, orderDate, orderTime, partNumber). The order key is needed to identify an order, and partNumber distinguishes its lines. Customer and part information depend on less than this full key, so the table is not in 2NF.

Question 1: Parts Order Form

2NF

Key notation: primary-key fields are **bold and underlined**.

Dependencies: (employee, orderDate, orderTime) determines customerNumber; customerNumber determines customerName and customerType; partNumber determines partName, partType, cageCode and unitPrice; the full line key determines quantityOrdered.

SalesOrder - 2NF

employee	**orderDate**	**orderTime**	customerNumber	customerName	customerType
D. Harrison	7/1/2024	10:30am	HG54587	Jeff Peterson	Consumer

Part - 2NF

partNumber	partName	partType	cageCode	unitPrice
10654	Float Control	Plumbing	G413	12
10456	Modulator	Electrical	H433	7
10776	Hose Assembly	Plumbing	G413	9
10657	Float Assembly	Plumbing	G413	10

OrderLine - 2NF

employee	**orderDate**	**orderTime**	**partNumber**	quantityOrdered
D. Harrison	7/1/2024	10:30am	10654	4
D. Harrison	7/1/2024	10:30am	10456	3
D. Harrison	7/1/2024	10:30am	10776	7
D. Harrison	7/1/2024	10:30am	10657	5

b. Explanation of the 2NF tables

The customer details for an order depend on (employee, orderDate, orderTime), while the part details depend on partNumber. Both depend on only part of the original primary key. I separate them into SalesOrder and Part to remove these partial dependencies. OrderLine keeps all four key fields and quantityOrdered, since the quantity depends on both the order and the part.

SalesOrder is in 2NF because its customer attributes depend on its entire order key, not a proper subset. However, it is not yet in 3NF: the order key determines customerNumber, which in turn determines customerName and customerType. Part and OrderLine already satisfy 3NF under the stated assumptions.

Question 1: Parts Order Form

3NF and keys

Key notation: primary-key fields are **bold and underlined**. Foreign-key fields are *italicized*; fields in both are ***bold, underlined and italicized***.

Customer - 3NF

customerNumber	customerName	customerType
HG54587	Jeff Peterson	Consumer

SalesOrder - 3NF

employee	**orderDate**	**orderTime**	*customerNumber*
D. Harrison	7/1/2024	10:30am	HG54587

Part - 3NF

partNumber	partName	partType	cageCode	unitPrice
10654	Float Control	Plumbing	G413	12
10456	Modulator	Electrical	H433	7
10776	Hose Assembly	Plumbing	G413	9
10657	Float Assembly	Plumbing	G413	10

OrderLine - 3NF

employee	***orderDate***	***orderTime***	***partNumber***	quantityOrdered
D. Harrison	7/1/2024	10:30am	10654	4
D. Harrison	7/1/2024	10:30am	10456	3
D. Harrison	7/1/2024	10:30am	10776	7
D. Harrison	7/1/2024	10:30am	10657	5

b. From 2NF to 3NF

In SalesOrder, the order key determines customerNumber, and customerNumber determines customerName and customerType. I move the customer details to Customer to remove this transitive dependency. SalesOrder keeps customerNumber as a foreign key. Part and OrderLine stay the same. The final tables have no partial or transitive dependencies.

c-d. Primary and foreign keys

Primary keys: Customer: customerNumber; SalesOrder: (employee, orderDate, orderTime); Part: partNumber; OrderLine: (employee, orderDate, orderTime, partNumber).

Foreign keys: SalesOrder.customerNumber references Customer.customerNumber. OrderLine.(employee, orderDate, orderTime) is **one composite foreign key** referencing the matching SalesOrder primary key. OrderLine.partNumber references Part.partNumber.

Question 2: Therapist Appointments

Original table and 1NF

Key notation: primary-key fields are **bold and underlined**.

TherapyRecord (1NF)

staffNo	therapistName	patNo	patName	**appointmentDate**	**appointmentTime**	branchNo
S1011	Fred Smith	P100	Lily White	9/12/2022	10:00	M15
S1011	Fred Smith	P105	Jill Baker	9/12/2022	12:00	M15
S1024	Heidi Pierce	P108	Andy McKee	9/12/2022	10:00	Q10
S1024	Heidi Pierce	P108	Andy McKee	9/14/2022	14:00	Q10
S1032	Richard Levin	P105	Jill Baker	9/14/2022	16:30	M15
S1032	Richard Levin	P110	Jimmy Winter	9/15/2022	18:00	B13

a. Assumptions

1. staffNo uniquely identifies a therapist and determines therapistName. patNo uniquely identifies a patient and determines patName. Names are not assumed unique.

2. A therapist has at most one appointment at a given date and start time; each appointment is with one patient. Therefore (staffNo, appointmentDate, appointmentTime) identifies one appointment. All key values are present.

3. As stated in the question, a therapist works at only one branch on any given day, but may work at different branches on different days. Thus (staffNo, appointmentDate) determines branchNo; staffNo alone does not.

4. A patient may have multiple appointments in a day and may see different therapists. Neither patNo alone nor (patNo, appointmentDate) is a suitable appointment key. No permanent patient-therapist or therapist-branch relationship is assumed.

b-c. 1NF and its primary key

The original table already has one appointment per row and one value per cell, so it is in 1NF. I label the date and time columns appointmentDate and appointmentTime. This only changes their names. The primary key is (staffNo, appointmentDate, appointmentTime).

It is not in 2NF: therapistName depends on staffNo alone, and branchNo depends on (staffNo, appointmentDate), both proper subsets of the primary key. Patient name also has a dependency through patNo that must be addressed by 3NF.

Question 2: Therapist Appointments

2NF

Key notation: primary-key fields are **bold and underlined**.

Therapist - 2NF

staffNo	therapistName
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard Levin

TherapistDay - 2NF

staffNo	**appointmentDate**	branchNo
S1011	9/12/2022	M15
S1024	9/12/2022	Q10
S1024	9/14/2022	Q10
S1032	9/14/2022	M15
S1032	9/15/2022	B13

Appointment - 2NF

staffNo	**appointmentDate**	**appointmentTime**	patNo	patName
S1011	9/12/2022	10:00	P100	Lily White
S1011	9/12/2022	12:00	P105	Jill Baker
S1024	9/12/2022	10:00	P108	Andy McKee
S1024	9/14/2022	14:00	P108	Andy McKee
S1032	9/14/2022	16:30	P105	Jill Baker
S1032	9/15/2022	18:00	P110	Jimmy Winter

b. Explanation of the 2NF tables

Therapist: therapistName depends only on staffNo, which is part of the original key. I place them in a separate table so the therapist's name is stored once.

TherapistDay: branchNo depends on staffNo and appointmentDate, but not on appointmentTime. I place these three fields in TherapistDay. Its key allows only one branch for each therapist on a given day, while allowing a different branch on another day.

Appointment: retain the complete appointment key plus patNo and patName. Under the assumptions, neither patient attribute depends on a proper subset of the key, so this relation is in 2NF. It still has a transitive dependency: the appointment key determines patNo, and patNo determines patName.

Remaining work: Therapist and TherapistDay already satisfy 3NF. Only Appointment needs another decomposition to remove the patient-name dependency.

Question 2: Therapist Appointments

3NF and keys

Key notation: primary-key fields are **bold and underlined**. Foreign-key fields are *italicized*; fields in both are ***bold, underlined and italicized***.

Therapist - 3NF

staffNo	therapistName
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard Levin

Patient - 3NF

patNo	patName
P100	Lily White
P105	Jill Baker
P108	Andy McKee
P110	Jimmy Winter

TherapistDay - 3NF

staffNo	**appointmentDate**	branchNo
S1011	9/12/2022	M15
S1024	9/12/2022	Q10
S1024	9/14/2022	Q10
S1032	9/14/2022	M15
S1032	9/15/2022	B13

Appointment - 3NF

staffNo	***appointmentDate***	**appointmentTime**	*patNo*
S1011	9/12/2022	10:00	P100
S1011	9/12/2022	12:00	P105
S1024	9/12/2022	10:00	P108
S1024	9/14/2022	14:00	P108
S1032	9/14/2022	16:30	P105
S1032	9/15/2022	18:00	P110

2NF to 3NF: move patName to Patient, keyed by patNo, and keep patNo in Appointment as a foreign key. Therapist and TherapistDay remain unchanged. Each remaining non-key attribute is determined directly by its relation's key, without a partial or transitive dependency.

Primary keys: Therapist: staffNo; Patient: patNo; TherapistDay: (staffNo, appointmentDate); Appointment: (staffNo, appointmentDate, appointmentTime).

Foreign keys: TherapistDay.staffNo references Therapist.staffNo. Appointment.(staffNo, appointmentDate) is **one composite foreign key** referencing TherapistDay.(staffNo, appointmentDate). Appointment.patNo references Patient.patNo.

Question 3: Event Staff

Original table and 1NF

Key notation: primary-key fields are **bold and underlined**.

StaffContractRecord (1NF)

eNo	**contractNo**	hours	eName	eventNo	eventLoc
1135	C1024	16	Smith J	H25	Queens
1057	C1024	24	Hocine D	H25	Queens
1068	C1025	28	White T	H4	Yonkers
1135	C1025	15	Smith J	H4	Yonkers
1135	C1026	10	Smith J	H25	Queens

a. Assumptions

1. eNo uniquely identifies an employee, as specified, and determines one recorded eName. Employee names are not assumed unique.

2. contractNo uniquely identifies a contract. Each contract applies to one event, while an event may have multiple contracts. eventNo uniquely identifies an event, and each event has one recorded location, eventLoc.

3. Each source row records the total hours worked by one employee under one contract. There is at most one aggregate row for each (eNo, contractNo) combination, and both values are present. Employees may work on several contracts, and contracts may involve several employees.

b-c. 1NF and its primary key

The source is already in 1NF: every cell contains one value and each row describes one employee-contract assignment. The primary key is (eNo, contractNo). eNo alone repeats across contracts, while contractNo alone repeats across employees; both are necessary.

Functional dependencies

eNo determines eName.
contractNo determines eventNo.
eventNo determines eventLoc.
(eNo, contractNo) determines hours.

This table is not in 2NF: eName depends only on eNo, while eventNo and, through eventNo, eventLoc depend only on contractNo. These are dependencies on proper subsets of the composite primary key.

Question 3: Event Staff

2NF

Key notation: primary-key fields are **bold and underlined**.

Employee - 2NF

eNo	eName
1135	Smith J
1057	Hocine D
1068	White T

Contract - 2NF

contractNo	eventNo	eventLoc
C1024	H25	Queens
C1025	H4	Yonkers
C1026	H25	Queens

WorkAssignment - 2NF

eNo	**contractNo**	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10

b. Explanation of the 2NF tables

Employee: move eName into a relation keyed by eNo. The employee name does not depend on which contract the employee works on, so it should not repeat in every work assignment.

Contract: move eventNo and eventLoc with contractNo. Both are determined by the contract portion of the original key, regardless of employee. Contract is now in 2NF because its key has only one attribute.

WorkAssignment: hours depends on both eNo and contractNo. I keep these fields together because the same employee can work different numbers of hours on different contracts.

Remaining problem: Contract is not yet in 3NF. contractNo determines eventNo, and eventNo determines eventLoc. Thus eventLoc has a transitive dependency on the Contract key. Employee and WorkAssignment already satisfy 3NF.

Question 3: Event Staff

3NF and keys

Key notation: primary-key fields are **bold and underlined**. Foreign-key fields are *italicized*; fields in both are ***bold, underlined and italicized***.

Employee - 3NF

eNo	eName
1135	Smith J
1057	Hocine D
1068	White T

Event - 3NF

eventNo	eventLoc
H25	Queens
H4	Yonkers

Contract - 3NF

contractNo	*eventNo*
C1024	H25
C1025	H4
C1026	H25

WorkAssignment - 3NF

eNo	***contractNo***	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10

b. From 2NF to 3NF

In Contract, contractNo determines eventNo, and eventNo determines eventLoc. I move eventLoc to Event to remove this transitive dependency. Contract keeps eventNo as a foreign key. For example, H25 has two contracts, C1024 and C1026, but its location is now stored only once. Employee and WorkAssignment stay the same.

c-d. Primary and foreign keys

Primary keys: Employee: eNo; Event: eventNo; Contract: contractNo; WorkAssignment: (eNo, contractNo).

Foreign keys: Contract.eventNo references Event.eventNo. WorkAssignment.eNo references Employee.eNo. WorkAssignment.contractNo references Contract.contractNo. The two foreign keys in WorkAssignment are separate, although together they form its primary key.

Each final non-key attribute depends on its entire primary key, with no partial or transitive dependency.